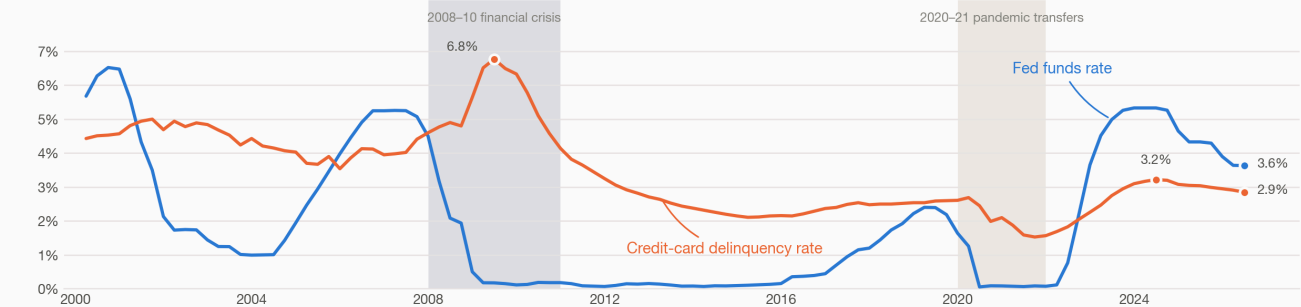


Rates and card delinquency: the two-year lag is one recession, not a rule

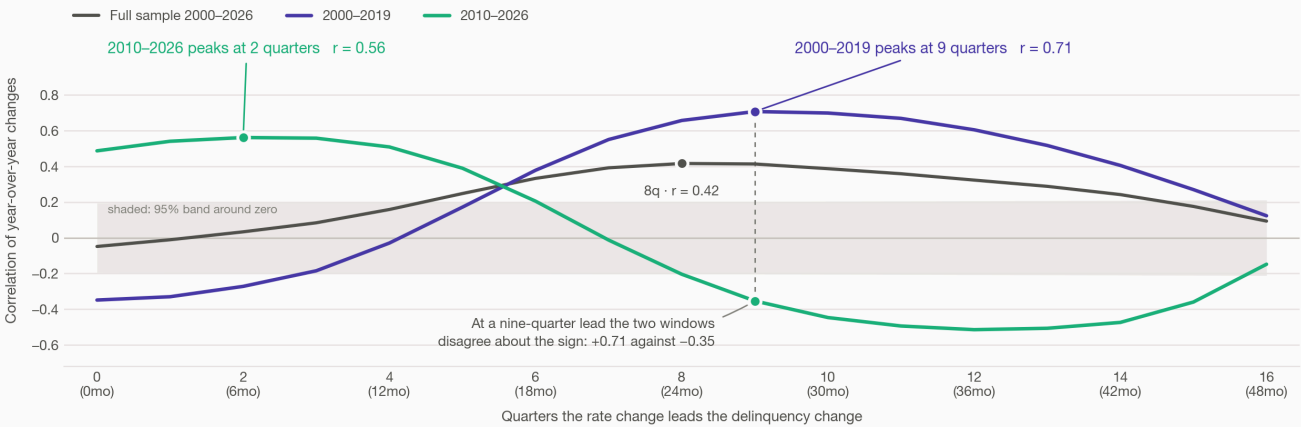
The two-year lag between rate hikes and card delinquency is one recession wearing a trend line

Twelve quarters around the 2008–10 crisis — 13% of the sample — supply 65% of the full-sample covariance at an eight-quarter lead. Drop those twelve quarters and the correlation falls from +0.42 to –0.07: the lead does not weaken, it disappears. The full-sample number is not a lag estimate but an average over two incompatible regimes, and a risk team forecasting on it would be reading the last crisis rather than this cycle.

Both series, one percent axis — the levels a risk officer already watches



Where the lead actually sits — and how far it moves when you change the window



Measured cycle by cycle — peak year-over-year rate rise to peak year-over-year delinquency rise — the three hiking cycles in this sample give three different answers: 2004–06 hiking, +14 quarters · 2015–18 hiking, –6 quarters (delinquency turned first) · 2022–23 hiking, +1 quarter.

Source: Federal Reserve Bank of St. Louis (FRED), FEDFUNDS, Federal Funds Effective Rate (monthly, averaged to quarters), through 2026-07-01, DRCCLACBS, Delinquency Rate on Credit Card Loans, All Commercial Banks (quarterly), through 2026-04-01; FRED last updated 2026-08-25, 106 aligned quarters, inner-joined. Changes are four-quarter differences. Retrieved 2026-08-27.

Fed funds rate and credit-card delinquency rate, 2000–2026, with cross-correlation of year-over-year changes by sample window

The finding

Run the standard analysis on the full sample and you get a clean, quotable answer: changes in the fed funds rate lead changes in credit-card delinquency by **eight quarters**, correlation **+0.42**. It looks like a planning parameter.

It is not one. Twelve quarters around the 2008–10 financial crisis — 13% of the sample — supply **65%** of the covariance behind that number. Remove those twelve quarters and the eight-quarter correlation

falls from **+0.42 to -0.07**. The lead does not weaken; it disappears. What the full sample is describing is one recession, and the two-year gap is the distance between the 2004–06 hiking cycle and the unemployment shock that actually drove the defaults.

Split the sample and the two halves contradict each other outright. On 2000–2019 the lead is 9 quarters ($r = +0.71$). On 2010–2026 it is 2 quarters ($r = +0.56$) — essentially contemporaneous. At a nine-quarter lead the two windows disagree about the **sign**: $+0.71$ against -0.35 . A team that fitted the full-sample rule and applied it to the current cycle would have had the direction wrong, not just the timing.

The mechanism is visible in the data. Unemployment moves with card delinquency at a **zero-quarter** lag in every pre-2020 window ($r = +0.58$ on 2000–2019) and more tightly than the policy rate does. In 2008 the rate was an early and coincidental marker of a labour-market collapse it did not cause. Since 2010, with no such shock in the sample, what remains is the fast channel — card APRs are indexed to prime, so a policy move reaches a household's minimum payment within a billing cycle or two, and delinquency responds inside a couple of quarters.

Results

Both series were pulled from FRED and aligned to quarters: FEDFUNDS is monthly and was averaged into each quarter (not sampled — a quarter's rate is the mean of its three months), DRCCLACBS is already quarterly, and the two were inner-joined, leaving **106 complete quarters from 2000 Q1 to 2026 Q2** with no gaps, no duplicates and no missing values.

Correlating the levels is not available: an ADF test cannot reject a unit root in the delinquency series ($p = 0.52$), so the $+0.21$ level correlation is spurious. All results below use **year-over-year (four-quarter) changes**, which are stationary for both series ($p = 0.003$ and $p = 0.030$) and are also the form a risk committee reads. A first-difference transform gives the same shape (peak at 9 quarters), so the conclusion is not an artefact of the transform.

Cross-correlation of year-over-year changes, by sample window:

SAMPLE	N	PEAK LEAD	R AT PEAK	R AT 0Q	R AT 9Q
Full, 2000–2026	94	8 quarters	+0.42	−0.05	+0.41
2000–2019	67	9 quarters	+0.71	−0.35	+0.71
2010–2026	64	2 quarters	+0.56	+0.49	−0.35
Full, less 2008–10	82	—	—	—	−0.07 at 8q

Effect sizes, from OLS of the delinquency change on the lagged rate change:

SAMPLE	LEAD	SLOPE PER +100BP	R ²
Full, 2000–2026	8q	+18bp (se 4)	0.17
2000–2019	9q	+37bp (se 5)	0.50
2010–2026	2q	+30bp (se 6)	0.32
2010–2026	8q	–7bp (se 5)	0.04

The last row is the cost of using the full-sample rule out of period: applied after 2010 it points the wrong way and explains nothing.

Two robustness checks. Leave-one-quarter-out on the full-sample eight-quarter correlation moves it only between +0.365 and +0.438 — so the result is not a single outlier quarter, it is a whole episode, which is the harder problem. And measured cycle by cycle, peak year-over-year rate rise to peak year-over-year delinquency rise, the three hiking cycles in the sample give three different answers: **2004–06, +14 quarters; 2015–18, –6 quarters** (delinquency turned first); **2022–23, +1 quarter**.

Where the current cycle stands: fed funds is **3.63%** and falling (–0.70pp year over year), card delinquency is **2.85%**, down from a cycle peak of 3.22% in 2024 Q2 and below the 2000–2026 average of 3.42%.

Source: Federal Reserve Bank of St. Louis (FRED). FEDFUNDS, Federal Funds Effective Rate, monthly, through 2026-07-01. DRCCLACBS, Delinquency Rate on Credit Card Loans, All Commercial Banks, quarterly, through 2026-04-01; FRED last updated 2026-08-25. Retrieved 2026-08-27. The delinquency series publishes about a quarter in arrears, so the joined sample ends one quarter before the rate data does.

What follows for a risk team

Do not carry a single lag number in a model or a policy document. The honest parameter is a range that moves with the regime: 1–2 quarters in the current one, up to 9 in a sample dominated by a credit crisis. If a scorecard, stress overlay or ECL model has a fixed rate-to-delinquency lag baked in, find out which window it was fitted on and when — a coefficient fitted on 2000–2019 is pointing backwards in this cycle.

Watch unemployment, not the policy rate, as the leading indicator. Rates lead delinquency only when they precede a labour-market shock, and then it is the shock doing the work. Unemployment is

contemporaneous with delinquency and strictly more informative. A rate-based early-warning trigger is really an untested bet that a downturn follows the hikes.

Size the rate channel from the current regime, and expect it to over-predict at large impulses. In the 2010–2026 window 100bp of policy maps to about 30bp of delinquency at a two-quarter lead, explaining roughly a third of the variance. Applied to this cycle that rule overshoots: the peak year-over-year rate impulse of +4.40pp implies +1.30pp of year-over-year delinquency against **+0.92pp observed**. Delinquency did rise 1.69pp from its 2021 Q3 trough to its 2024 Q2 peak, but part of that was normalisation from a pandemic low held down by fiscal transfers. A linear rate coefficient is a serviceable first-order term and a poor stress parameter — at large impulses it reads high.

Re-estimate on a rolling window, and publish the instability. This relationship changed twice in twenty-six years. Any model that assumes it is fixed will look accurate in backtest and fail in the next regime. Re-fit on a rolling window, monitor whether the peak lead has moved, and report the range rather than the point estimate — the honest deliverable to a risk committee is the fact that this parameter is unstable, not a false precision about where it currently sits.

One caveat on scope. This is two public aggregate series for all commercial banks. It says nothing about a specific bank's mix, vintage, credit tier or underwriting, and a subprime-weighted portfolio may well have a different and faster response. Treat it as a prior about the macro channel and a caution about fixed lag assumptions — not as a substitute for fitting the relationship on our own portfolio.

FinTechCo is fictional and this estate is a stand-in — no production system and no real customer data. The analysis uses public FRED series; the chart is the run's own output.